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USN | | | | | | | | | | [23CSE108]

# FIRST SEMESTER B.Tech EXAMINATION, JANUARY -2025

## FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS ENGINEERING

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

| | | | Blooms Level | CO | Marks |
|---|---|---|---|---|---|
| | | **PART-A** | **5 x2=10** | | |
| **1.** | **a.** | State and explain Ohm's Law and its limitations. | L2 | CO1 | **02** |
| | **b.** | Define the following:<br>**i.** Electric circuit<br>**ii.** Electrical circuit elements | L2 | CO2 | **02** |
| | **c.** | Explain briefly the necessity of transformer. | L2 | CO3 | **02** |
| | **d.** | Write the diode current equation. | L2 | CO4 | **02** |
| | **e.** | What is a rectifier? | L2 | CO5 | **02** |
| | | | | | |
| | | **PART-B** | **6 X 5=30** | | |
| **2.** | **a.** | List the differences between Active and Passive components. | L2 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | State Kirchhoff's Current and Voltage Law. | L2 | CO1 | **05** |
| | | | | | |
| **3.** | **a.** | Sketch the sinusoidal alternating wave form and define amplitude, time period, Instantaneous Value. | L3 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | Explain the concept of phase and phase difference in alternating current. | L2 | CO2 | **05** |
| | | | | | |
| **4.** | **a.** | Describe the principle of operation of a single-phase transformer. | L2 | CO3 | **05** |
| | | **OR** | | | |
| | **b.** | Discuss the different types of losses in a transformer and their causes. | L2 | CO3 | **05** |
| | | | | | |
| **5.** | **a.** | Explain the mechanism of avalanche breakdown in a PN junction diode. | L2 | CO4 | **05** |
| | | **OR** | | | |
| | **b.** | Compare the operation of NPN and PNP transistors. | L2 | CO4 | **05** |
| | | | | | |
| **6.** | **a.** | Explain the working of a half-wave rectifier with a circuit diagram. | L2 | CO5 | **05** |
| | | **OR** | | | |

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| | **b.** | Explain with a neat sketch the working of simple Zener Diode as voltage regulator. | L3 | CO5 | **05** |
| | | | | | |
| **7.** | **a.** | List the differences between Alternating Current and Direct Current. | L2 | CO1 | 05 |
| | | **OR** | | | |
| | **b.** | List the differences between Resistor, Inductor and Capacitor. | L2 | CO1 | 05 |
| | | | | | |
| | | **PART-C** | **6X10=60** | | |
| **8.** | **a.** | Explain ideal and practical current sources. | L2 | CO1 | **10** |
| | | **OR** | | | |
| | **b.** | Explain ideal and practical voltage sources. | L2 | CO1 | **10** |
| | | | | | |
| **9.** | **a.** | Define average value and derive an expression for the average value of a sinusoidal current. | L3 | CO2 | **10** |
| | | **OR** | | | |
| | **b.** | Define RMS or Effective Value and derive an expression for the RMS or Effective value of sinusoidal current. | L3 | CO2 | **10** |
| | | | | | |
| **10.** | **a.** | Describe the construction and types of single-phase transformers. | L2 | CO3 | **10** |
| | | **OR** | | | |
| | **b.** | Sketch and explain the V-I characteristics of a PN junction diode under forward and reverse bias. | L3 | CO3 | **10** |
| | | | | | |
| **11.** | **a.** | Illustrate the principle operation of a Common Emitter (CE) NPN transistor with a diagram. | L2 | CO4 | **10** |
| | | **OR** | | | |
| | **b.** | Illustrate the principle operation of a Common Base (CB) NPN transistor with a diagram. | L2 | CO4 | **10** |
| | | | | | |
| **12.** | **a.** | Explain the structure, operation, and working principle of PNP and NPN transistors. Support your explanation with labeled diagrams. | L2 | CO5 | **10** |
| | | **OR** | | | |
| | **b.** | Compare the construction, working, and applications of NPN and PNP transistors in detail. Include diagrams and circuit examples. | L2 | CO5 | **10** |
| | | | | | |
| **13.** | **a.** | Explain the working principles of full-wave bridge rectifiers. Include circuit diagrams. | L2 | CO6 | **10** |
| | | **OR** | | | |
| | **b.** | Describe the full wave rectification process and analyze the efficiency. | L2 | CO6 | **10** |